

Q1 - 25 January - Shift 2

If m and n respectively are the numbers of positive and negative value of θ in the interval $[-\pi, \pi]$ that satisfy the equation $\cos 2\theta \cos \frac{\theta}{2} = \cos 3\theta \cos \frac{9\theta}{2}$, then mn is equal to _____.

Space for your notes:

Q2 - 29 January - Shift 1

Let $f(\theta) = 3 \left(\sin^4 \left(\frac{3\pi}{2} - \theta \right) + \sin^4 (3\pi + \theta) \right) - 2(1 - \sin^2 2\theta)$ and

Space for your notes:

$$S = \left\{ \theta \in [0, \pi] : f'(\theta) = -\frac{\sqrt{3}}{2} \right\}. \text{ If } 4\beta = \sum_{\theta \in S} \theta,$$

then $f(\beta)$ is equal to

(1) $\frac{11}{8}$

(2) $\frac{5}{4}$

(3) $\frac{9}{8}$

(4) $\frac{3}{2}$

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Answer Key

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(As per Official NTA Key released on 2 Feb)

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Q1 (25)

Q2 (2)

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Q1 (25)

$$\cos 2\theta \cdot \cos \frac{\theta}{2} = \cos 3\theta \cdot \cos \frac{\theta}{2}$$

$$\Rightarrow 2 \cos 2\theta \cdot \cos \frac{\theta}{2} = 2 \cos \frac{9\theta}{2} \cdot \cos 3\theta$$

$$\Rightarrow \cos \frac{5\theta}{2} + \cos \frac{3\theta}{2} = \cos \frac{15\theta}{2} + \cos \frac{3\theta}{2}$$

$$\Rightarrow \cos \frac{15\theta}{2} = \cos \frac{5\theta}{2}$$

$$\Rightarrow \frac{15\theta}{2} = 2k\pi \pm \frac{5\theta}{2}$$

$$5\theta = 2k\pi \text{ or } 10\theta = 2k\pi$$

$$\theta = \frac{2k\pi}{5} \quad \theta = \frac{k\pi}{5}$$

$$\therefore \theta = \left\{ -\pi, \frac{-4\pi}{5}, \frac{-3\pi}{5}, \frac{-2\pi}{5}, \frac{-\pi}{5}, 0, \frac{\pi}{5}, \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{4\pi}{5}, \pi \right\}$$

$$m = 5, n = 5$$

$$\therefore m.n = 25$$

Q2 (2)

$$f(\theta) = 3 \left(\sin^4 \left(\frac{3\pi}{2} - \theta \right) + \sin^4 (3\pi + \theta) \right) - 2(1 - \sin^2 2\theta)$$

$$S = \left\{ \theta \in [0, \pi] : f'(\theta) = -\frac{\sqrt{3}}{2} \right\}$$

$$\Rightarrow f(\theta) = 3(\cos^4 \theta + \sin^4 \theta) - 2\cos^2 2\theta$$

$$\Rightarrow f(\theta) = 3 \left(1 - \frac{1}{2} \sin^2 2\theta \right) - 2\cos^2 2\theta$$

$$\Rightarrow f(\theta) = 3 - \frac{3}{2} \sin^2 2\theta - 2\cos^2 2\theta$$

$$= \frac{3}{2} - \frac{1}{2} \cos^2 2\theta = \frac{3}{2} - \frac{1}{2} \left(\frac{1 + \cos 4\theta}{2} \right)$$

$$f(\theta) = \frac{5}{4} - \frac{\cos 4\theta}{4}$$

$$f'(\theta) = \sin 4\theta$$

$$\Rightarrow f'(\theta) = \sin 4\theta = -\frac{\sqrt{3}}{2}$$

$$\Rightarrow 4\theta = n\pi + (-1)^n \frac{\pi}{3}$$

$$\Rightarrow \theta = \frac{n\pi}{4} + (-1)^n \frac{\pi}{12}$$

$$\Rightarrow \theta = \frac{\pi}{12}, \left(\frac{\pi}{4} - \frac{\pi}{12} \right), \left(\frac{\pi}{2} + \frac{\pi}{12} \right), \left(\frac{3\pi}{4} - \frac{\pi}{12} \right)$$

$$\Rightarrow 4\beta = \frac{\pi}{4} + \frac{\pi}{2} + \frac{3\pi}{4} = \frac{3\pi}{2}$$

$$\Rightarrow \beta = \frac{3\pi}{8} \Rightarrow f(\beta) = \frac{5}{4} - \frac{\cos \frac{3\pi}{2}}{4} = \frac{5}{4}$$

Questions with Solutions

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